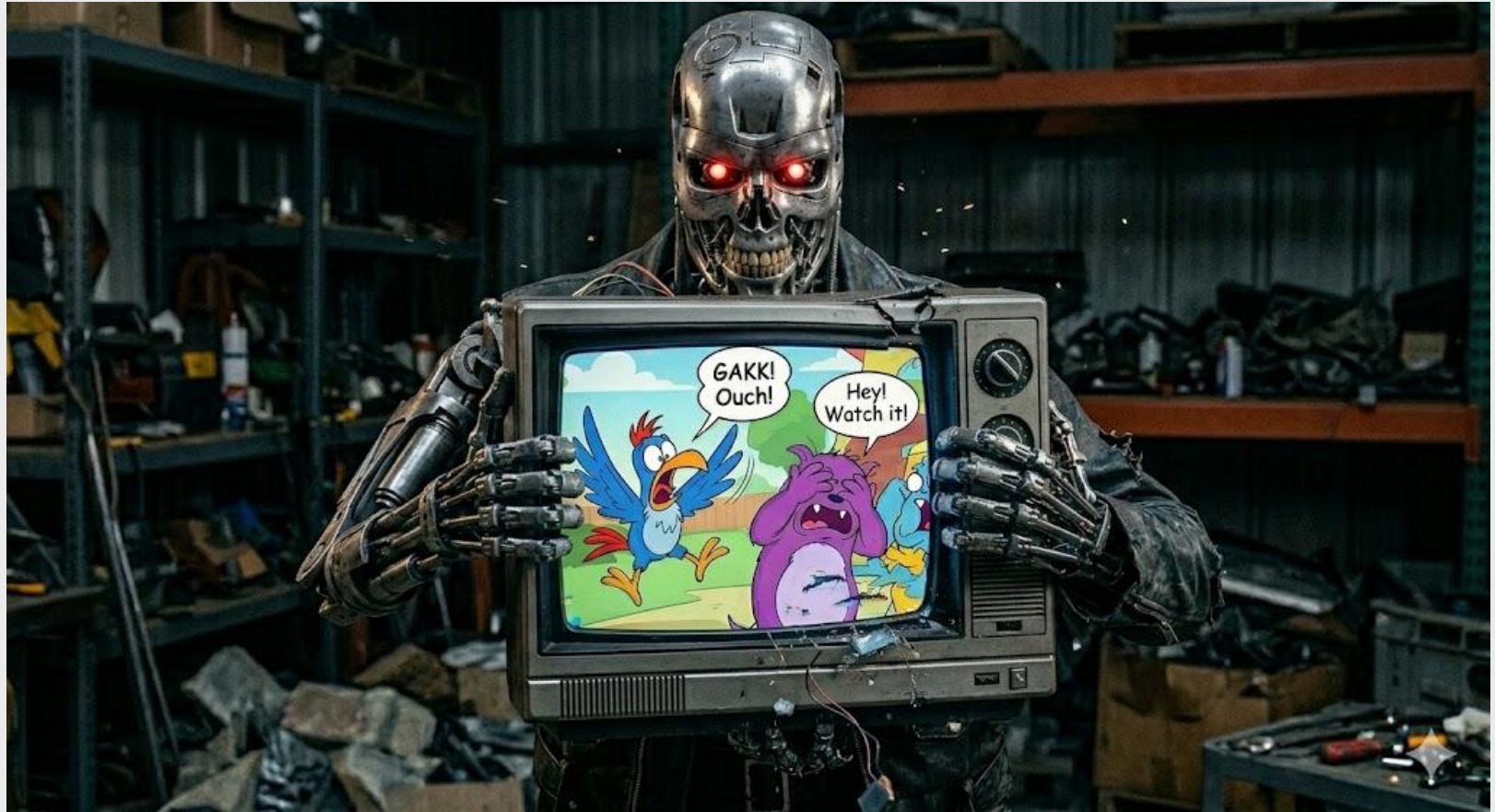


AI Video Summarization

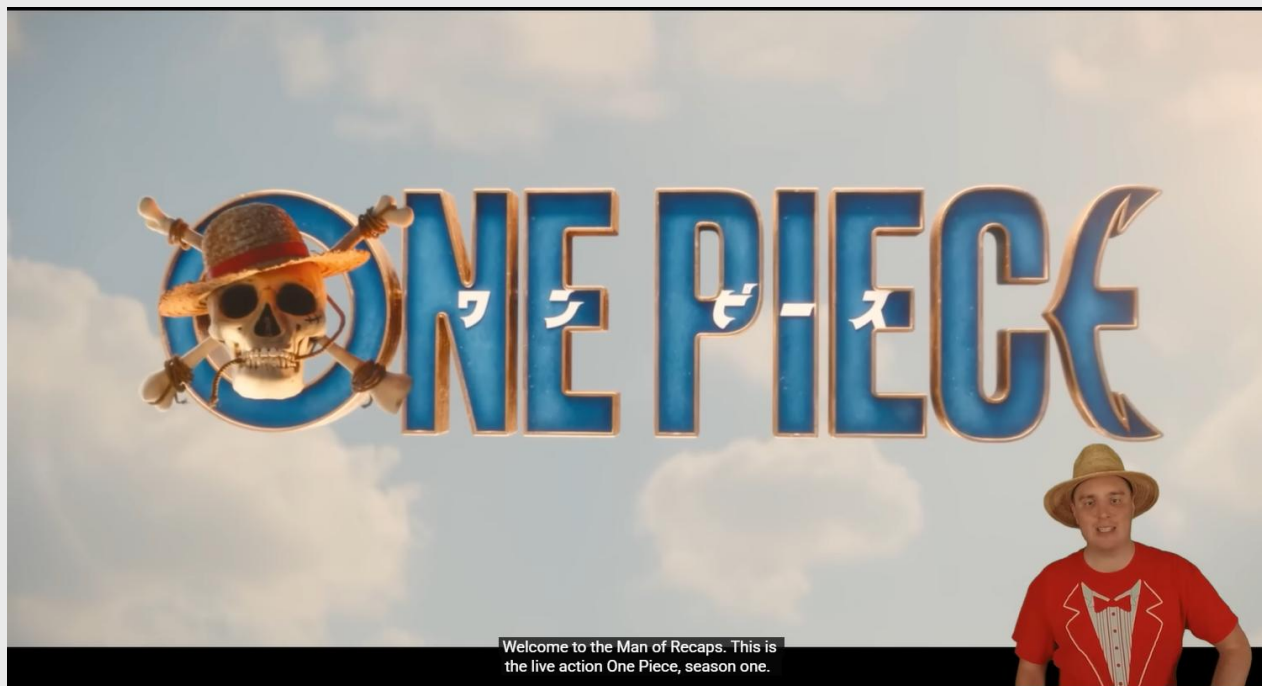


Why Summarize Video?

- **Time – turn long-form content (films, lectures, sports) into short watchable summaries (movie trailers, high-light reels, etc)**
- **Targeting – highlight elements of a video that resonate with a particular audience**
- **Discovery – help viewers decide what to watch**
- **Education – lecture and documentary clips for study guides**

Example: Man of Recaps

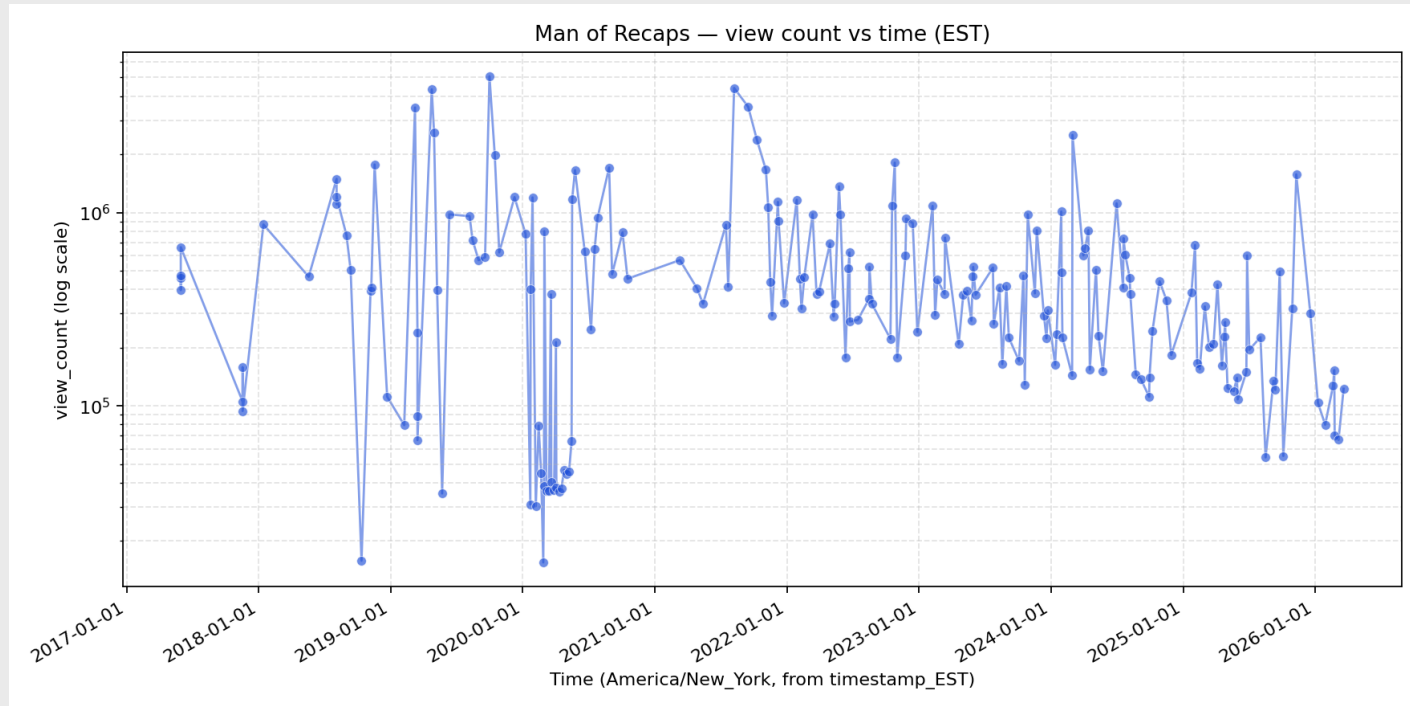
- Hosted recap video - cold open, episode or movie title, then voiceover over stitched clips from the source
- Similar channels on YouTube use this pattern for TV, anime, and film—search for recap or “story explained” formats.



Summarization Channel

 Peak Blinders RECAP 18:45	 Outlander RECAP 29:28	 A Knight of the Seven Kingdoms Season 1 RECAP 8:00	 Paradise Season 1 RECAP 9:45
Peak Blinders RECAP: Full Series before The Immortal Man 123K views • 10 days ago	Outlander RECAP before the Final Season 66K views • 3 weeks ago	A Knight of the Seven Kingdoms RECAP: Season 1 70K views • 1 month ago	Paradise RECAP: Season 1 153K views • 1 month ago
 The Night Agent Season 2 RECAP 6:53	 Bridgerton Season 3 RECAP 6:52	 The Pitt Season 1 RECAP 9:29	 Avatar The Way of Water RECAP 5:44
The Night Agent RECAP: Season 2 127K views • 1 month ago	Bridgerton RECAP: Season 3 79K views • 2 months ago	The Pitt RECAP: Season 1 103K views • 2 months ago	Avatar RECAP: The Way of Water 300K views • 3 months ago
 Stranger Things RECAP 18:13	 The Witcher Seasons 1 - 3 RECAP 11:22	 Harry Potter Bedtime Story 22:14	 Alice in Borderland RECAP 13:41
Stranger Things RECAP before the Final Season 1.5M views • 4 months ago	The Witcher RECAP before Season 4 318K views • 5 months ago	Fall Asleep to Harry Potter as a Bedtime Story 55K views • 5 months ago	Alice in Borderland RECAP before Season 3 494K views • 6 months ago

Channel Views



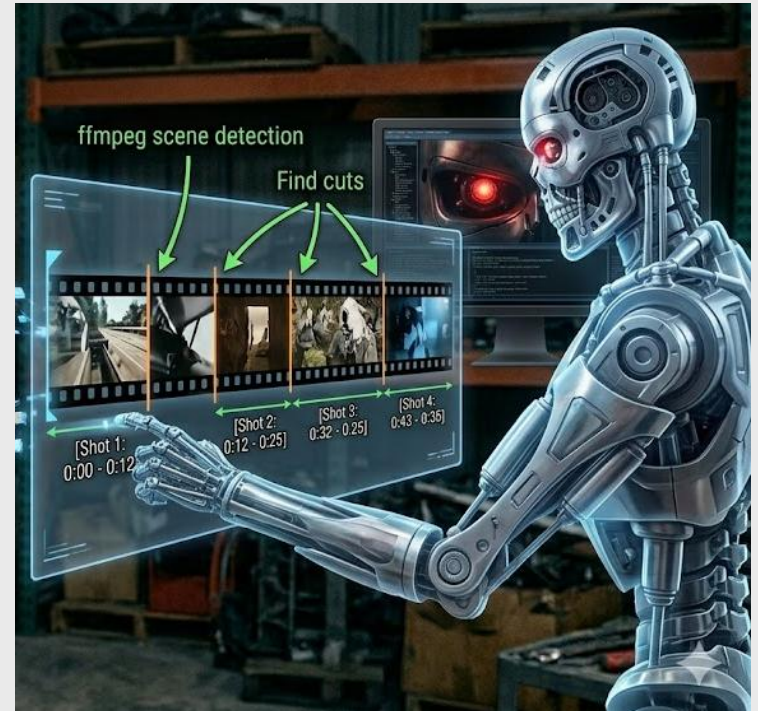
- **202 videos over 9 years**
- **Views per video:**
 - mean = 581K,
 - median = 378K,
 - max = 5M

Raw Inputs

- To summarize a video with AI we need two elements:
video file and **subtitle file**
- Video file – full episode or movie (e.g. mp4, mkv)
 - We will sample frames from this file to get a visual understanding of the shots
- Subtitle file – srt file with time-stamped dialogue
 - Can be extracted with ffmpeg library from video file sometimes
 - *Ask your AI vibe coder to download and install ffmpeg for you*
- Time-stamped dialogue + visual samples give enough signal for AI to describe each shot

Shot Detection

- Use ffmpeg scene detection (or similar) to find cuts: each interval is a shot with start/end times on the source timeline.
- Output: a manifest (e.g. shots.json) — timing only; hundreds of short shots for a feature film is normal.
- Parameters:
 - SCENE_THRESHOLD – lower bound on how different 2 frames need to be to declare a new shot
 - MIN_GAP_SECONDS – minimum duration of a shot



Example of shots.json

```
{
  "video_path":
    "C:\\Users\\tr2\\Dropbox\\ChatGPTs\\mgt575_apps\\2026Spring\\lecture_16_video_summarizat
    ion\\test\\video_raw\\Back.to.the.Future.Part.2.1989.1080p.Brrip.x264.Deceit.YIFY.mp4",
  "video_duration_seconds": 6472.4267,
  "scene_detection": {
    "scene_threshold": 0.35,
    "min_gap_seconds": 10,
    "num_detected_cuts": 157
  },
  "num_shots": 158,
  "shots": [
    {
      "index": 0,
      "start_seconds": 0.0,
      "end_seconds": 36.5417,
      "duration_seconds": 36.5417
    },
    {
      "index": 1,
      "start_seconds": 36.5417,
      "end_seconds": 47.7917,
      "duration_seconds": 11.25
    },
  ],
}
```

Building a Shot Catalogue

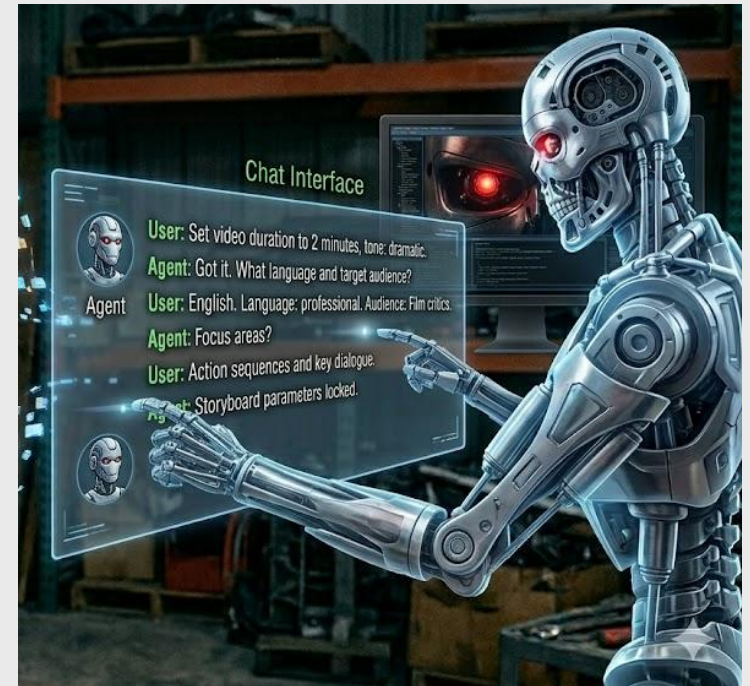
- For each shot: sample several **frames + subtitle lines + previous shot descriptions** → send to AI
- Output: shots_description.json — shot index, times, dialogue excerpt and AI generated description per shot
- Approx. 15 minutes for 150 shots

Example of shots_description.json

```
{
  "index": 86,
  "start_seconds": 1986.8333,
  "end_seconds": 2008.375,
  "description": "Lorraine, Marty's future mother, prepares a pizza using a futuristic  
\"Black & Decker Hydrator\" appliance. She unwraps a tiny, foil-packaged pizza,  
places it on a tray into the machine, and it rapidly expands into a full-sized pizza.  
",
  "dialogue": "Hey, pizza. I'm hungry. All right. Just wait your turn. Grandma, when  
it's ready, could you just shove it in my mouth? Don't you be a smartass. Oh, great!  
The Atrocity Channel. Hydrate level 4, please."
},
{
  "index": 87,
  "start_seconds": 2008.375,
  "end_seconds": 2018.8333,
  "description": "Lorraine serves the freshly hydrated pizza to her family, including  
older Marty, Marty Jr., and Marlene, who are gathered around the dinner table.  
Meanwhile, the younger Jennifer (from 1985) secretly observes the family dinner from  
a doorway.",
  "dialogue": "Mmm. Is it ready? Here you go. Oh, boy. Oh, boy. Mom, you sure can  
hydrate a pizza."
},
{
```

Storyboard Agent

- We use an agent with a chat interface to make our summary video storyboard
- Give agent the shot catalogue, a video style template (optional) and chat with agent to get video parameters (duration, tone, language, target audience, things to focus on)
- Output storyboard.json – ordered scenes with time stamps on source video and narration text for scenes



Example of storyboard.json

```
"scenes": [  
  {  
    "index": 0,  
    "start_time": 36.5417,  
    "stop_time": 47.7917,  
    "narration": "[smooth] Welcome to the Man of Recaps. [upbeat] Today we will be  
    covering Back to the Future Part 2. [energetic] Get ready for a wild ride through  
    time!",  
    "source_shot_indices": [  
      1  
    ]  
  },  
  {  
    "index": 1,  
    "start_time": 78.2083,  
    "stop_time": 86.0,  
    "narration": "[urgent] Doc Brown bursts in, telling Marty: \"You've got to come back  
    with me!\"",  
    "source_shot_indices": [  
      4  
    ]  
  },  
]
```

Render Video

- **For each storyboard scene, extract the source video clip (ffmpeg)**
- **TTS (e.g. ElevenLabs) reads the narration**
 - emotion tags like [excited] steer delivery.
- **Mux audio with video**
 - adjust playback speed of video to match voiceover length
 - speed up or slow down video so no awkward silences
- **Concatenate scenes into final summary video**

Coding Session

- Make an agent to create AI video summaries
- Recaps, trailers, highlights

